

Multi-Behavior Hypergraph Contrastive Learning for Session-Based Recommendation

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Abstract—Most current session-based recommendations model session sequences solely based on the user’s target behavior, ignoring the user’s hidden preferences in auxiliary behaviors. Additionally, they use ordinary graphs to model one-to-one item correlations in the current session and fail to leverage other sessions to learn richer higher-order item correlations. To address these issues, a multi-behavior hypergraph contrastive learning model for session-based recommendations is proposed. This model represents all the sessions as global hypergraphs according to two types of behavior sequences. It employs contrastive learning to obtain global item embeddings, which are further aggregated to generate a global session representation that captures higher-order correlations of items from all session perspectives. A novel local heterogeneous hypergraph is designed for the current session to capture higher-order correlations between items with different behaviors in the current session, thus enhancing the local session representation. Additionally, a novel self-supervised signal is created by constructing a multi-behavior line graph, enhancing the global session representation. Finally, the local session representation, global session representation, and global item embedding are used to learn the predicted interaction probability of each item. Extensive experiments are conducted on three real datasets, and the results demonstrate that the proposed model significantly improves recommendation accuracy.

Index Terms—Contrastive learning, hypergraph, line graph, multi-behavior, session-based recommendation.

I. INTRODUCTION

TRADITIONAL recommendation systems predominantly assume the availability of user profiles and a long history of user interactions, making recommendations on the basis of these data. However, in the real world, the user’s identity and long history of interactions may be unknown. In this scenario, only interaction records from the current session can be used for recommendation. Therefore, session-based recommendations have received increasing attention in recent years. The basic

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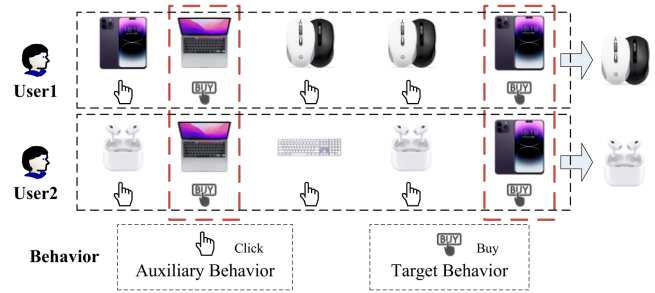


Fig. 1. Impact of items with different auxiliary behaviors on an item with the next target behavior.

form of session-based recommendation is to learn vectors that effectively represent user intent on the basis of the current session. For sparse session data, relying solely on item IDs for modeling is insufficient to fully capture user intentions. Therefore, exploiting auxiliary information beyond item IDs (e.g., user interaction behaviors with items) to enhance user intention representation has become an important branch of session-based recommendation research.

In e-commerce, user purchasing behavior generates direct value for the merchant. Thus, part of the research on session-based recommendations [1], [2], [3], [4] aims to predict the next most likely interaction item on the basis of the user’s target behavior (i.e., purchasing behavior). There are many types of interactions between users and items, such as clicking, adding items to shopping carts, and purchasing. Some studies [52], [53], [54] have used multiple behaviors to improve recommendation performance. As illustrated in Fig. 1, users 1 and 2 may have the same purchase items (e.g., laptops and mobile phones), leading the system to recommend the same items to them, regardless of the auxiliary behavior (clicking). However, in reality, the two users might choose different items (users 1 and 2 might choose a mouse and a headset, respectively). Therefore, when modeling only the target behavior, the rich user preferences hidden in the auxiliary behaviors are ignored. Nevertheless, most existing studies of session-based recommendation [5], [6], [7] use only a single interaction behavior for recommendations, with only a few researchers [3], [8], [9], [10], [11] considering the role of auxiliary behaviors (e.g., adding to a shopping cart, clicking) outside of the target behavior.

In addition, most studies [12], [13], [14] model only one-to-one correlations between items, making it difficult to fully utilize item interaction information. In the real world, the interaction with an item may be driven by a combination of multiple

previous items with various interaction behaviors. This suggests a higher-order correlation between items, involving many-to-many correlations. For example, a purchase may be driven by multiple clicks or purchases, or by multiple clicks and purchases together; that is, items within the same behavior and between different behaviors have higher-order correlations. Therefore, to utilize interaction information fully, it is necessary to model the higher-order correlations of items in multi-behavioral model when introducing auxiliary behaviors into session recommendation. Thus, this study addresses the following two major challenges.

- 1) Modeling higher-order correlations of items within the same behavior and between different behaviors simultaneously in a single graph from all session perspectives is complex and difficult to learn. Therefore, the first challenge is to use all sessions separately to model higher-order correlations of items within the same behavior and between different behaviors, thereby better leveraging all sessions for richer user intentions.
- 2) Modeling only from all session perspectives can easily be misled by popular item interactions, which necessitates the extraction of user preferences from a single session (i.e., the current session). However, when modeling with the current session, most studies have modeled target and auxiliary behaviors separately and individually, disrupting the relative positions of items with different interaction behaviors in the original session sequence. Clicking on the same item before and after purchase clearly indicates different user intentions. In addition, different interaction behaviors with an item and different users for the same type of behavior indicate different user preferences. Therefore, the second challenge is to model the higher-order correlations of items between different behaviors in a single session and to embody user preferences for different behaviors to fully exploit user intentions in the current session.

To address the first challenge, we learn global item embeddings by constructing global hypergraphs from the target behavior sequences and auxiliary behavior sequences of all sessions to capture higher-order correlations of items within the same behavior. To obtain more comprehensive global item embeddings, we employ contrastive learning, which enables the global item embeddings learned from the global hypergraphs on the basis of target and auxiliary behaviors to augment each other. To capture the higher-order correlations of items between different behaviors from all session perspectives, we design a novel self-supervised signal (i.e., a line graph session representation learned from a multi-behavioral line graph) for the global session representation. The multi-behavior line graph includes two types of edges: those related to sequences of target and auxiliary behaviors, which can be good for interbehavioral information transfer during the line graph convolution process.

For the second challenge, on the one hand, we constructed a local session heterogeneous hypergraph for the current session to capture higher-order correlations of items between different interaction behaviors while preserving the relative positions between items with different interaction behaviors. Items with

auxiliary behaviors may themselves be noise (e.g., items clicked aimlessly). Therefore, the sequence of items with auxiliary behaviors in the current session does not need to follow a strict temporal order. This is better represented by the hyperedges in the hypergraphs. On the other hand, personalized behavioral weights are designed in the graph convolution process to capture the user's personalized preferences for different behaviors. When generating the local session representation, the soft-attention mechanism with two different query vectors is designed to aggregate the items in the current session from the perspective of general preference and current preference. Thus, the influence of noise in the auxiliary behavior on the results is reduced to a certain extent, resulting in a more accurate extraction of the user intentions from the current session.

We propose a novel multi-behavior hypergraph contrastive learning method for session-based recommendation (MHCL). The main contributions of this study are as follows:

- 1) Based on all the sessions, global item embeddings are learned that incorporate the higher-order correlations of the items in two types of behavioral sequences to provide a reliable input for subsequent components of the proposed model. In addition, a multi-behavior line graph is designed to further model the higher-order correlations of items between different behaviors from a global session perspective, effectively augmenting the final global session representation by contrastive learning.
- 2) A multi-behavior-based local session heterogeneous hypergraph was constructed. While preserving the relative positions of items with different behaviors in the original session, the higher-order correlations of items between different behaviors in the current session were fully mined. Moreover, personalized behavior weights are designed for users, and the items are aggregated from two attention perspectives to reduce the influence of noise in auxiliary behaviors and obtain a more accurate local session representation.
- 3) Multiple baseline models were compared on three datasets, and extensive experimental results demonstrate that MHCL can effectively improve the accuracy of session recommendations.

II. RELATED WORK

A. Session-Based Recommendation

As recurrent neural networks (RNNs) have advantages in modeling sequential patterns, several studies [15], [16], [17] have utilized RNNs to model user intentions in sessions. Although RNN-based models show some improvement in the effectiveness of session-based recommendations, they focus primarily on modeling the sequential correlations of adjacent items and cannot capture complex item correlations. Graph neural networks (GNNs) are widely used in session-based recommendations to better model item correlations in sessions because of their ability to capture more complex item correlations. Several studies [18], [19], [20], [21] construct each session as a directed or undirected graph, learning one-to-one correlations between items via a GNN to extract item features. However, these models

only learn the correlations between items in the current session, which is insufficient for fully mining item features and user intentions, especially for sparse session sequences.

Several researchers have investigated the extraction of valuable insights from sessions beyond the current session to enhance the representation of the current session. Several studies [22], [23], [24] have obtained more accurate session representations by selecting relevant information from pertinent sessions within all sessions for modeling. Some other studies [12], [13], [14] capture item correlations from both the current session and all sessions, allowing for the learning of popular-biased and personalized-biased item representations, respectively. Han et al. [25] devised a multifaceted global item relation (MGIR), extracted globally incompatible item relations from all sessions and fused sequential, cooccurring, and incompatible item relations to improve recommendation performance. Li et al. [26] used a disentangled technique to explore users' fine-grained preferences and generate session representations. Wang et al. [27] modeled user preferences from both individual- and group-level session perspectives to accurately predict the repeated consumption of users. However, most of the above studies only modeled the low-order correlations of items.

Because hypergraphs provide the ability to model higher-order correlations of items, some studies have begun to model sessions as hypergraphs. Dual-channel hypergraph convolutional network (S²-DHCN) [29] are the first approach to use hypergraphs in the field of session-based recommendation. This approach performs two-channel learning of hypergraphs and line graphs, enhancing session representations using contrastive learning. Several studies [30], [31] analyze possible user preference transitions, construct personalized hyperedges, and capture higher-order correlations of different granularities in the hypergraph. Some studies have modeled heterogeneous item attributes as hypergraphs to enhance session representation. Ma et al. [32] designed a category-aware lossless heterogeneous hypergraph neural network (CLHHN) and modeled the attribute information of items into a heterogeneous hypergraph. Zhang et al. [4], [33] constructed a heterogeneous hypergraph network that emphasizes the importance of price factors in influencing users' purchasing behavior. However, these studies consider only the target behavior and ignore the auxiliary behavior.

Auxiliary behaviors, in addition to target behaviors, have been introduced [3], [10], [11], [34], [35], [36], [37], [38], [39], [51] to enhance session representation. Several studies [3], [10], [11], [34], [35], [37], [39] have typically separated sessions into target behavior sequences and auxiliary behavior sequences, modeling them separately. This approach disrupts the item correlations between the target and auxiliary behaviors and fails to adequately capture their mutual influence. The global and personalized graphs for heterogeneous sequential recommendation (GPG4HSR) [36] and heterogeneous information crossing on graphs [38] use ordinary heterogeneous graphs, which overemphasize the sequential relationships between items within a behavior and can only show one-to-one correlations of items.

Therefore, in this study, we consider both target and auxiliary behaviors and combine all the sessions with the current session to

model the sessions with multiple behaviors. This is done by constructing global hypergraphs and heterogeneous hypergraphs to capture more complex higher-order correlations of items.

B. Contrastive Learning

Contrastive learning is a self-supervised learning method that generates positive and negative samples using a custom rule designed to learn feature representations by maximizing the similarity between positive samples and minimizing the similarity between negative samples. Owing to its great success in fields such as graphics and vision [40], [41], [42], several studies [43], [44], [45], [46] have introduced contrastive learning in sequential recommendations. However, these studies use a randomized exit strategy for data augmentation to generate contrastive signals, which is not applicable to session-based recommendations. This is because the randomized exit strategy makes the already sparse session data even sparser. S²-DHCN and a self-supervised graph cotraining framework for session-based recommendations (COTREC) [47] utilize an enhancement strategy without a random exit to construct two graphs that supervise each other through contrastive learning for better data enhancement. Pan et al. [48] constructed a global graph to establish connections between different sessions and used contrastive learning to augment item feature representation, taking each session's long-term preferences and its neighbors on the global graph as self-supervised signals for contrastive learning. Peintner et al. [49] applied self-supervised contrastive learning to mine positive and negative item samples in a data-driven manner, thereby mitigating the risks of data sparsity and popularity bias. Wan et al. [50] used spatiotemporal contrastive learning to extract self-supervised signals from spatial and temporal views to mitigate the loss of temporal information and improve the quality of feature learning.

However, these contrastive learning methods have been applied only to session-based recommendations for a single behavior. In this study, we used contrastive learning to effectively fuse the information extracted from auxiliary behaviors with that extracted from target behaviors. Additionally, we adopted a nonrandom exit strategy to design a self-supervised signal for contrastive learning for global session representation, which is more suitable for session-based recommendation scenarios with sparse data.

III. PROPOSED MODEL

This section first formally defines the problem to be solved and then details the proposed MHCL.

A. Problem Definition

Suppose that a set of items V is $\{v_1, v_2, \dots, v_N\}$, where N denotes the number of items, and a set of behavior types B is $\{b_1, b_2\}$, where b_1 is the target behavior and where b_2 is the auxiliary behavior. The input to the model in this paper is a set of anonymous sessions $\mathcal{S} = \{s_1, s_2, \dots, s_J\}$, representing all the sessions, where J is the number of sessions in the set. The current session $s \in \mathcal{S}$ is a sequence of interacting items



B. MHCL

1) *Initial Item Embedding*: In this study, we use one-hot encoding to represent the input item IDs, obtaining a one-hot encoding representation $\mathbf{v}_i \in \mathbb{R}^N$ of each item v_i . Furthermore, a learnable embedding matrix $\mathbf{W}_1 \in \mathbb{R}^{N \times d}$, where d is the embedding dimension) is used to convert the one-hot representation to a low-dimensional dense vector representation. The initial embedding $\mathbf{h}_i \in \mathbb{R}^{1 \times d}$ of item v_i is obtained as shown in (1):

$$h_i = \boldsymbol{v}_i^T \mathbf{W}_1 \quad (1)$$

where the initial values of the elements in $\mathbf{W}_1$ are random values uniformly distributed in the range of $[-d^{-1/2}, d^{-1/2}]$, which are trained by the *Adam* optimizer in the backpropagation process. $\boldsymbol{\nu}_i^T$ denotes the transpose of $\boldsymbol{\nu}_i$.

2) *Global Item Embedding*: To utilize item higher-order correlations from all session perspective, we consider the natural advantage of hypergraph modeling of higher-order correlation and construct a global hypergraph based on the target behavior using the target behavior sequences of all sessions to learn global item embeddings based on target behavior. To learn valuable information from the auxiliary behavior sequence for item embedding, the auxiliary behavior sequences of all the sessions are used to obtain the global item embedding based on the auxiliary behavior in the same manner as above. The two types of global item embeddings are used as contrastive signals to acquire information from each other through contrastive learning, resulting in augmented global item embeddings based on the target behavior (hereinafter referred to as “augmented global item embeddings”). These augmented global item embeddings incorporate higher-order correlations of items hidden in all session target behaviors and auxiliary behaviors, which are then used as inputs for subsequent components.

Next, taking the learning of global item embeddings on the basis of target behavior as an example, the corresponding method is expounded.

a) *Global hypergraph construction based on target behavior*: The global hypergraph based on the target behavior is defined as $G_q^b = (V, E_q^b)$. All the items in V are nodes of the

global hypergraph, and $E_g^{b_1}$ is the set of hyperedges. The number of hyperedges (denoted as J) is the number of sessions in the set $\mathcal{S}$, where each session corresponds to a single hyperedge. Any hyperedge $e_{g:j}^{b_1} (\in E_g^{b_1})$ in $G_g^{b_1}$ typically contains multiple nodes, where each node represents a target behavior item in the session s_j ($j = 1, 2, \dots, J$). For hypergraph $G_g^{b_1}$, the adjacency relation is represented by a matrix $\mathbf{P} (\in R^{N \times J})$, where the element in row i and column j is p_{ij} . If the hyperedge $e_{g:j}^{b_1}$ contains item v_i , p_{ij} is 1; otherwise, it is zero.

b) *Global hypergraph convolution based on target behavior*: Global hypergraph convolution can be viewed as the process of information propagation along node-hyperedge-node, with the specific convolution operation defined as

$$\mathbf{h}_i^{(l)} = \sum_{j=1}^J \sum_{k=1}^N p_{ij} p_{kj} \mathbf{h}_k^{(l-1)} \quad (2)$$

where $\mathbf{h}_i^{(l)}$ denotes the embedding of item v_i in layer l ($l = 1, 2, \dots, L_g$), $\mathbf{h}_k^{(l-1)}$ represents the embedding of item v_k in layer $l-1$, and $\mathbf{h}_i^{(0)} = \mathbf{h}_i$ represents the initial embedding of item v_i .

The embedding of each item v_i needs to be processed by L_g layers of global hypergraph convolution. To prevent low-order information from being lost during the convolution process, the embeddings obtained from each layer as well as the initial embedding of the item are subjected to a mean pooling operation to obtain its global item embedding $\mathbf{h}_{g:i}^{b_1} (\in R^{1 \times d})$ on the basis of target behavior in the following manner:

$$\mathbf{h}_{g:i}^{b_1} = \frac{1}{L_g + 1} \sum_{l=0}^{L_g} \mathbf{h}_i^{(l)} \quad (3)$$

Similarly, a global hypergraph $G_g^{b_2}$ ($G_g^{b_2} = (V, E_g^{b_2})$) is constructed for the auxiliary behavior sequences of all sessions. Following the same global hypergraph convolution operation described above, the global item embedding $\mathbf{h}_{g:i}^{b_2}$ based on auxiliary behavior for item v_i is derived from the global hypergraph based on auxiliary behavior.

c) *Contrastive learning for global item embedding*: Different global embeddings, $\mathbf{h}_{g:i}^{b_1}$ and $\mathbf{h}_{g:i}^{b_2}$, learned from different hypergraphs for the same item v_i are used to complement each other through contrastive learning. They are regarded as truth labels for each other. This encourages the global item embeddings based on target behavior and the global item embeddings based on auxiliary behavior representing the same item to be similar, resulting in augmented global item embeddings. Similar to [29], the standard binary cross-entropy loss between the positive and negative samples is used as the objective of contrastive learning, as shown in (4):

$$\mathcal{L}_{item} = -\log \sigma \left(f \left(\mathbf{h}_{g:i}^{b_1}, \mathbf{h}_{g:i}^{b_2} \right) \right) - \log \sigma \left(1 - f \left(\tilde{\mathbf{h}}_{g:i}^{b_1}, \mathbf{h}_{g:i}^{b_2} \right) \right) \quad (4)$$

where $\tilde{\mathbf{h}}_{g:i}^{b_1}$ denotes the negative sample obtained by row and column transformations of the matrix $\mathbf{H}_g^{b_1}$, $\mathbf{H}_g^{b_1}$ represents the matrix consisting of augmented global item embeddings $\mathbf{h}_{g:i}^{b_1} (i = 1, 2, \dots, N)$ of all items, i serves as the row index, $f(\cdot) : R^d \times R^d \mapsto R$ is a discriminator function to score the

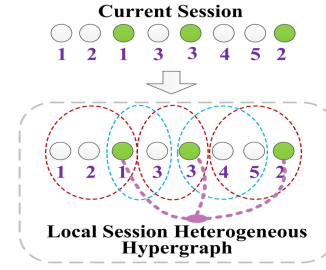


Fig. 3. Local session heterogeneous hypergraph.

consistency of the two embeddings of the input, and $\sigma(\cdot)$ is the Sigmoid function:

3) *Local Session Representation*: The goal of the module is to capture the higher-order correlations of items between behaviors under the current session and accurately learn user-personalized local item embeddings. Thus, accurate local session representations are obtained, minimizing the noise that is introduced in auxiliary behaviors. Therefore, we construct a local session heterogeneous hypergraph for the current session to preserve the relative positions between the target and auxiliary behavior items in the original current session, thus capturing higher-order correlations between the target behavior and auxiliary behavior items more effectively. The process of local session representation learning is as follows: First, the local session heterogeneous hypergraph is constructed using the current session s . The L_s -layer heterogeneous hypergraph convolution operation is subsequently performed on the heterogeneous hypergraph. Finally, the local session representation is obtained by embedding aggregation via the readout function.

a) *Local session heterogeneous hypergraph construction*: We define the local session heterogeneous hypergraph $G_{loc} = (B, V_{loc}, E_{loc})$, where V_{loc} is the set of nodes. Each node corresponds to an item, and E_{loc} is the set of hyperedges, where the number of hyperedges J_{loc} is determined by the number of target behavior items in s such that $J_{loc} = 2x_{b_1} + 1$. The heterogeneous hypergraph is shown in Fig. 3 (the items are arranged in the graph in order of timestamp from smallest to largest, from left to right, and 1, 2, 3, 4, and 5 denote the item IDs). All the items in the current session are treated as nodes in the hypergraph. The target behavior items serve as boundaries to form different hyperedges with the preceding and following auxiliary behavior items. A target behavior item forms a hyperedge (shown by the red dashed ellipse) with auxiliary behavior items that have timestamps smaller than the target behavior item's timestamp and larger than the previous target behavior item's timestamp, as well as a hyperedge (shown by the blue dashed ellipse) with auxiliary behavior items that have timestamps larger than the target behavior item's timestamp and smaller than the next target behavior item's timestamp. Additionally, all target behavior items are constructed as a hyperedge (shown by thick curves connected by elliptical dots). In this way, the target behavior item and its preceding and following auxiliary behavior items are included in two different hyperedges, and the auxiliary behavior item and its preceding and following target behavior items are also included in two different hyperedges, preserving

the relative positions between the target behavior item and the auxiliary behavior item.

The local session heterogeneous hypergraph is represented by the matrix $\mathbf{Q} \in R^{2 \times J_{loc} \times x}$. If item v_{t_τ} (its interaction behavior is b_{t_τ}) belongs to the hyperedge $e_{loc:j}$, then the element $q_{kj\tau} = 1$ in the matrix $\mathbf{Q}$; otherwise, $q_{kj\tau} = 0$ (if b_{t_τ} is the target behavior, then $k = 1$; otherwise, $k = 2$). The hyperedge $e_{loc:j} (\in E_{loc})$ is the j th hyperedge in G_{loc} . The first target behavior item in s forms the first hyperedge with the preceding auxiliary behavior items, and the second hyperedge with the group of auxiliary behavior items immediately following it. In this order, all target behavior items form the J_{loc} th hyperedge.

b) Local session heterogeneous hypergraph convolution: The information propagation process of a local session heterogeneous hypergraph still uses node-hyperedge-node propagation. In this process, the items in the local session heterogeneous hypergraph share information with other items in the same hyperedge to which they belong. This approach allows for the efficient extraction of higher-order correlations of items within the hyperedge. Additionally, the higher-order correlations of items in neighboring hyperedges are also fully explored after multiple rounds of information propagation. When a node propagates information to a hyperedge, each interaction behavior is assigned a corresponding weight to distinguish the importance of different interaction behaviors. Because different users have different operating habits, we cannot simply assign the same interaction behavior weights to all users. We obtain the behavior weights by calculating the similarity of the items under the auxiliary and target behaviors in each session. This similarity indicates the frequency with which the user performs auxiliary behaviors on the same or similar items before performing the target behavior. When the difference between the user's auxiliary behavior item and the target behavior item is too large, the auxiliary behavior is considered to contain more noise, and its weight will be smaller, which can effectively reduce the influence of noise on the results.

Specifically, for the current session, the augmented global item embeddings of the items under different interaction behaviors are subjected to a mean pooling operation to obtain the corresponding behavior mean vector $\mathbf{u}^{b_k} = \frac{1}{x_{b_k}} \sum_{(v_{t_\tau}, b_{t_\tau}) \in s \text{ and } b_{t_\tau} = b_k} \mathbf{h}_{g:t_\tau}^{b_1}$ ($k = 1 \text{ or } 2$). Here, $\mathbf{h}_{g:t_\tau}^{b_1}$ denotes the augmented global item embedding of item v_{t_τ} in the current session. The similarity between the different behavior mean vectors and the target behavior mean vector is computed using the vector dot product, and a normalization process is performed to obtain the behavior weights ω^{b_k} ($= \text{softmax}(\mathbf{u}^{b_k} \cdot \mathbf{u}^{b_1})$) of the nodes in the current session when they are aggregated toward the hyperedge, where ω^{b_1} and ω^{b_2} denote the target behavior weights and auxiliary behavior weights, respectively. The node-to-hyperedge aggregation operation is expressed as follows:

$$\boldsymbol{\varepsilon}_j^{(l)} = \sum_{k=1}^2 \sum_{\tau=1}^x \omega^{b_k} \times q_{kj\tau} \times \mathbf{h}_{s:\tau}^{(l-1)} \quad (5)$$

where $\boldsymbol{\varepsilon}_j^{(l)} (\in R^{1 \times d})$ denotes the embedding of the hyperedge $e_{loc:j}$ at layer l ($l = 1, \dots, L_s$), and where $\mathbf{h}_{s:\tau}^{(l-1)}$ is the local

item embedding of item v_{t_τ} at layer $l - 1$ in the current session s . In particular, $\mathbf{h}_{s:\tau}^{(0)}$ is the augmented global item embedding of item v_{t_τ} , that is, $\mathbf{h}_{g:v_{t_\tau}}^{b_1}$.

Different hyperedges in the local session heterogeneous hypergraph contain relative position information between different behavioral items, and different user preferences are hidden in the position information. We distinguish the different effects of different relative location information on user preferences by distinguishing the importance of different hyperedges. Therefore, in the process of hyperedge-to-node aggregation, we use a soft-attention mechanism to distinguish the importance of different hyperedges on the basis of the local item embedding of each item in the previous layer as the query vector. Specifically, for the hyperedge $e_{loc:j}$ ($j \in \{1, 2, \dots, J_{loc}\}$ and $q_{kj\tau} = 1$) containing item v_{t_τ} (the corresponding behavior is $b_{t_\tau} = b_k$), its importance, i.e., the attention coefficient, can be obtained via (6):

$$\xi_j^{(l)} = \text{softmax} \left(\left(\boldsymbol{\varepsilon}_j^{(l)} \mathbf{W}_1^e \right) \times \left(\mathbf{h}_{s:\tau}^{(l-1)} \mathbf{W}_2^e \right)^T / \sqrt{d} \right) \quad (6)$$

where $\xi_j^{(l)}$ is the attention coefficient of $e_{loc:j}$ for item v_{t_τ} at layer l and where $\mathbf{W}_1^e (\in R^{d \times d})$ and $\mathbf{W}_2^e (\in R^{d \times d})$ are trainable parameter matrices. We aggregate the hyperedge embeddings using the attention coefficients to generate the local item embedding of item v_{t_τ} in the current session s at layer l as $\mathbf{h}_{s:\tau}^{(l)}$:

$$\mathbf{h}_{s:\tau}^{(l)} = \sum_{j \in \{1, 2, \dots, J_{loc}\} \text{ and } q_{kj\tau} = 1} \xi_j^{(l)} \boldsymbol{\varepsilon}_j^{(l)} \quad (7)$$

To retain more information from the upper layers and prevent the loss of information owing to the increase in the number of layers during the convolution of the local heterogeneous hypergraph, we fuse the local item embeddings of layer l with those of layer $l - 1$ through an addition operation, obtaining the fused local item embedding $\bar{\mathbf{h}}_{s:\tau}^{(l)}$, i.e., $\bar{\mathbf{h}}_{s:\tau}^{(l)} = \mathbf{h}_{s:\tau}^{(l)} + \mathbf{h}_{s:\tau}^{(l-1)}$. Finally, for the fused local item embedding $\bar{\mathbf{h}}_{s:\tau}^{(l)}$ of each layer, a mean pooling operation is performed to obtain the final local item embedding $\mathbf{h}_{s:\tau}$ of item v_{t_τ} :

$$\mathbf{h}_{s:\tau} = \frac{1}{L_s + 1} \left(\sum_{l=1}^{L_s} \bar{\mathbf{h}}_{s:\tau}^{(l)} + \bar{\mathbf{h}}_{s:\tau}^{(0)} \right) \quad (8)$$

where $\bar{\mathbf{h}}_{s:\tau}^{(0)} = \mathbf{h}_{s:\tau}^{(0)}$.

c) Local item embedding aggregation: To extract user preferences from the local item embedding of the current session s , we use the soft-attention mechanism to aggregate local item embeddings. In addition, to extract user preferences from different perspectives, we use two different query vectors for the soft-attention mechanism. These vectors are oriented toward the hyperedge $e_{loc:J_{loc}}$ containing all target behavior items in the current session and the hyperedge $e_{loc:J_{loc}-1}$ containing the last target behavior item and its subsequent auxiliary behavior items in the current session. The former reflects the user's general preferences, whereas the latter better reflects the user's current preferences. On the basis of these two query vectors, different session representations are generated by the soft-attention mechanism. The query vector used for general preferences contains only

the target behavior items to effectively reduce the noise in the auxiliary behavior items. A method similar to (5) is used for the aggregation of nodes to hyperedges, obtaining the query vectors $\vartheta_{J_{loc}-1} = \sum_{k=1}^2 \sum_{\tau=1}^x \omega^{b_k} \times q_{k(J_{loc}-1)\tau} \times \mathbf{h}_{s:\tau}$ and $\vartheta_{J_{loc}} = \sum_{k=1}^2 \sum_{\tau=1}^x \omega^{b_k} \times q_{kJ_{loc}\tau} \times \mathbf{h}_{s:\tau}$ for hyperedges $e_{loc:J_{loc}-1}$ and $e_{loc:J_{loc}}$. The soft-attention mechanism is as follows:

$$\beta_s^\beta = \sum_{\tau=1}^x \beta_\tau \mathbf{h}_{s:\tau} \quad (9)$$

$$\zeta_s^\zeta = \sum_{\tau=1}^x \zeta_\tau \mathbf{h}_{s:\tau} \quad (10)$$

$$\beta_\tau = \sigma(\mathbf{h}_{s:\tau} \mathbf{W}_1^\beta + \vartheta_{J_{loc}-1} \mathbf{W}_2^\beta) \mathbf{W}_3^\beta \quad (11)$$

$$\zeta_\tau = \sigma(\mathbf{h}_{s:\tau} \mathbf{W}_1^\zeta + \vartheta_{J_{loc}} \mathbf{W}_2^\zeta) \mathbf{W}_3^\zeta \quad (12)$$

where β_τ and ζ_τ denote the attention coefficients of item v_{t_τ} under the two query vectors $\vartheta_{J_{loc}-1}$ and $\vartheta_{J_{loc}}$, respectively, $\sigma(\cdot)$ represents the *Sigmoid* function, and $\mathbf{W}_1^\beta, \mathbf{W}_2^\beta, \mathbf{W}_1^\zeta$ and $\mathbf{W}_2^\zeta$ (all $\in R^{d \times d}$), $\mathbf{W}_3^\beta$ and $\mathbf{W}_3^\zeta$ (both $\in R^{d \times 1}$) stand for the trainable parameter matrices. These attention coefficients are used as weights to aggregate the local item embeddings of different items, generating the current-preference-oriented session representation $\mathbf{z}_s^\beta$ and the general-preference-oriented session representation $\mathbf{z}_s^\zeta$ for the current session s .

Finally, the two session representations are aggregated using a gating mechanism to obtain a more comprehensive local session representation, $\mathbf{z}_s$.

$$\mathbf{z}_s = \alpha \mathbf{z}_s^\beta + (1 - \alpha) \mathbf{z}_s^\zeta \quad (13)$$

where α denotes the gating coefficient and $\alpha = \sigma(\mathbf{z}_s^\beta \mathbf{W}_1^z + \mathbf{z}_s^\zeta \mathbf{W}_2^z)$, $\sigma(\cdot)$ represents the *Sigmoid* function, and $\mathbf{W}_1^z \in R^{d \times 1}$ and $\mathbf{W}_2^z \in R^{d \times 1}$ indicate trainable parameter matrices.

4) *Global Session Representation*: First, the soft-attention mechanism with positional information was utilized to obtain the global session representation on the basis of the target behavior sequence and the global session representation on the basis of the auxiliary behavior sequence. To obtain a unified global session representation, the global session representations for both behaviors were then fused using a mean pooling operation to reduce the number of parameters when generating the global session representation. To extract the higher-order correlations of items between different behaviors from the global perspective and obtain a more comprehensive global session representation, a multi-behavior line graph was constructed on the basis of the global hypergraph, and a line graph session representation was generated through convolution operations on the multi-behavior line graph, thus providing session representation from another perspective. Finally, contrastive learning was used to enhance the global session representation.

a) *Global session representation generation*: For the target behavior sequence s^{b_1} in the current session s , the augmented global item embedding for each item in ascending timestamp order is $[\mathbf{h}_{g:t(1)}^{b_1}, \mathbf{h}_{g:t(2)}^{b_1}, \dots, \mathbf{h}_{g:t(x_{b_1})}^{b_1}]$. $\mathbf{h}_{g:t(i)}^{b_1}$ is the augmented global item embedding for the i th ($i = 1, 2, \dots, x_{b_1}$) item in s^{b_1} . Then, a learnable positional embedding matrix, $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_{x_{b_1}}]$, is used to add a reverse positional embedding for each item in s^{b_1} . $\mathbf{h}_{g:t(i)}^{b_1}$

($= \tanh([\mathbf{h}_{g:t(i)}^{b_1} || \mathbf{m}_{x_{b_1}-i}] \mathbf{W}^m)$) represents the item embedding of the i th item in s^{b_1} after adding the position information. $\mathbf{W}^m \in R^{2d \times d}$ denotes the trainable parameter matrix.

Subsequently, item embeddings with positional information are aggregated via a soft-attention mechanism to obtain a global session representation $\mathbf{z}_{g(s)}^{b_1}$ of session s on the basis of the target behavior:

$$\mathbf{z}_{g(s)}^{b_1} = \sum_{i=1}^{x_{b_1}} \eta_i \mathbf{h}_{g:t(i)}^{b_1} \quad (14)$$

$$\eta_i = \sigma(\mathbf{h}_{g:t(i)}^{b_1} \mathbf{W}_1^{b_1} + \bar{\mathbf{h}} \mathbf{W}_2^{b_1}) \mathbf{W}_3^{b_1} \quad (15)$$

where η_i denotes the attention coefficient of the item at the corresponding position; $\sigma(\cdot)$ denotes the *Sigmoid* function; $\mathbf{W}_1^{b_1}$ and $\mathbf{W}_2^{b_1}$ (both $\in R^{d \times d}$) and $\mathbf{W}_3^{b_1}$ ($\in R^{d \times 1}$) are trainable parameter matrices; and $\bar{\mathbf{h}}$ is the average of all the above item embeddings with positional information, i.e., $(\sum_{i=1}^{x_{b_1}} \mathbf{h}_{g:t(i)}^{b_1}) / x_{b_1}$, as the query vector for the soft-attention mechanism.

For the auxiliary behavior sequence, we use operations similar to those described above to generate the corresponding global session representation $\mathbf{z}_{g(s)}^{b_2}$ on the basis of the auxiliary behavior. The only difference is that the position embedding is removed.

Finally, the global session representations based on the target and auxiliary behaviors are aggregated via a mean pooling operation to generate a comprehensive global session representation $\mathbf{z}_{g(s)}$ for session s :

$$\mathbf{z}_{g(s)} = \frac{1}{2} \sum_{k=1}^2 \mathbf{z}_{g(s)}^{b_k} \quad (16)$$

b) *Session contrastive learning*: First, drawing on the approach in the literature [29] and introducing auxiliary behavior information to expand it into a multi-behavior model, a multi-behavior line graph was constructed on the basis of the global hypergraphs of different behaviors using the sessions in the current training batch. The local session representations (containing higher-order correlations of items in the current session and user preferences from a single session perspective) obtained in Section III-B3 are then used as the initial embeddings of nodes in the multi-behavior line graph. After multilayered line graph convolutions, we can efficiently transfer the local session representation and enhance the line graph session representation. Finally, in conjunction with the above global session representation, contrastive learning is performed to enhance the global session representation. This is expressed as follows:

Multi-behavior line graph construction: The multi-behavior line graph G_{lin} is a heterogeneous ordinary graph constructed by extracting key information from global hypergraphs. Each node in the line graph represents a session (the number of sessions in the current training batch is the number of nodes in the graph), and the edges represent the connectivity information between sessions. Here, we use undirected edges. In G_{lin} , two types of edges are set, and each type of edge is extracted from the global hypergraph of the corresponding behavior. For example, for the global hypergraph $G_g^{b_1}$, each hyperedge corresponds to a node in graph G_{lin} . If two hyperedges in $G_g^{b_1}$ have common nodes, then there is an edge connection between the two

corresponding nodes in G_{lin} . For the global hypergraph $G_g^{b_2}$, the above situation is similar. Thus, the multi-behavior line graph can be formally defined as $G_{lin} = (V_{lin}, E_{lin})$, where the set of nodes $V_{lin} = \{v_{ei} : v_{ei} \in E_g^{b_k}\}$, and the set of edges $E_{lin} = \{(v_{ei}, v_{ej})_k : |e_{g:i}^{b_k} \cap e_{g:j}^{b_k}| \geq 1\}$, $(v_{ei}, v_{ej})_k$ ($k \in \{1, 2\}$) denote the k th type of edges in the multi-behavior line graph, $E_g^{b_k}$ and $e_{g:i}^{b_k}$ are the set of hyperedges of the global hypergraph based on the behavior b_k and the i th hyperedge in it, respectively, and $|e_{g:i}^{b_k} \cap e_{g:j}^{b_k}|$ is the number of common nodes contained in the two hyperedges. In addition, a weight $a_{kij} = |e_{g:i}^{b_k} \cap e_{g:j}^{b_k}| / |e_{g:i}^{b_k} \cup e_{g:j}^{b_k}|$ is set for each edge $(v_{ei}, v_{ej})_k$ to indicate the degree of association between two nodes in the multi-behavior line graph, where $|e_{g:i}^{b_k} \cup e_{g:j}^{b_k}|$ is the total number of nodes contained in the two hyperedges (with repeated nodes counted only once).

Multi-behavior line graph convolution: A multi-behavior line graph contains connectivity information between sessions in a global hypergraph. If there are edge connections between nodes, the nodes are defined as neighbors of each other, and the convolution of the multi-behavior graph is the process of aggregating the neighbor information of a node to the node. On the basis of the multi-behavior line graph G_{lin} constructed above, we define the matrix $\mathbf{A} \in R^{2 \times \mathcal{Y} \times \mathcal{Y}}$ ($\mathcal{Y}$ ($\mathcal{Y} < J$) is the number of nodes, i.e., the number of sessions in the current batch during the training process), whose element is the weight a_{kij} of each edge in the line graph. To avoid nodes losing their own information during the convolution process, the association matrix $\hat{\mathbf{A}}_k \in R^{\mathcal{Y} \times \mathcal{Y}}$, and $k = 1, 2$) is defined ($\hat{\mathbf{A}}_k = \mathbf{A}_k + \mathbf{I}$, where $\mathbf{I}$ is the identity matrix for the purpose of adding self-connecting edges of the nodes, and $\mathbf{A}_k$ is the matrix in which $\mathbf{A}$ is indexed to k in the first dimension) and adds self-connecting edges for each node in the line graph. We define the degree matrix $\hat{\mathbf{D}}_k \in R^{\mathcal{Y} \times \mathcal{Y}}$ as a diagonal matrix with elements $\hat{d}_{kii} = \sum_{j=1}^{\mathcal{Y}} \hat{a}_{kij}$ in $\hat{\mathbf{D}}_k$, and $\hat{a}_{kij}$ is an element in $\hat{\mathbf{A}}_k$. The multi-behavior line graph convolution is defined as follows:

$$\boldsymbol{\theta}^{(l)} = (\hat{\mathbf{D}}_1)^{-1} \hat{\mathbf{A}}_1 \boldsymbol{\theta}^{(l-1)} + \mu (\hat{\mathbf{D}}_2)^{-1} \hat{\mathbf{A}}_2 \boldsymbol{\theta}^{(l-1)} \quad (17)$$

where $\boldsymbol{\theta}^{(l)}$ and $\boldsymbol{\theta}^{(l-1)}$ denote the multi-behavior line graph node embeddings in the l th ($l = 1, 2, \dots, L_g$) and $(l-1)$ th layers, respectively. Specifically, $\boldsymbol{\theta}^{(0)} = \mathbf{z}_s$. μ is the hyperparameter that controls the extent to which the neighbors of a node in the line graph aggregate information to the node from the edges of the auxiliary behavior type. $(\cdot)^{-1}$ denotes the inverse operation of the matrix.

After the above convolution process, the node embeddings $\boldsymbol{\theta}^{(l)}$ at each layer are averaged to obtain the corresponding line graph session representation $\boldsymbol{\theta}_{(s)}$ for the current session s : $\boldsymbol{\theta}_{(s)} = (\sum_{l=0}^{L_g} \boldsymbol{\theta}^{(l)}) / (L_g + 1)$.

Creating self-supervised signals: In sessions trained with the same batch, the global session representation and line graph session representation representing the same session are considered positively correlated samples; otherwise, they are labeled negatively correlated samples. Specifically, if session s_j is trained in the current batch, then its global session representation $\mathbf{z}_{g(s_j)}$

and line graph session representation $\boldsymbol{\theta}_{(s_j)}$ are positively correlated samples to each other. The matrix consisting of the global session representations (or line graph session representations) of all the sessions in the current training batch undergoes row and column transformations, and the vector at the original index position of session s_j is taken to obtain the negative correlation samples $\tilde{\mathbf{z}}_{g(s_j)}$ (or $\tilde{\boldsymbol{\theta}}_{(s_j)}$) of $\boldsymbol{\theta}_{(s_j)}$ (or $\mathbf{z}_{g(s_j)}$). Therefore, the standard binary cross-entropy loss between the positive and negative samples is used as the learning objective and is defined as

$$\mathcal{L}_{lin} = -\log \sigma \left(f(\mathbf{z}_{g(s_j)}, \boldsymbol{\theta}_{(s_j)}) \right) - \log \sigma \left(1 - f(\tilde{\mathbf{z}}_{g(s_j)}, \tilde{\boldsymbol{\theta}}_{(s_j)}) \right) \quad (18)$$

where $\sigma(\cdot)$ denotes the *Sigmoid* function and $f(\cdot) : R^d \times R^d \mapsto R$ represents a discriminator function (similar to that in Section III-B2)). The global session representation can be effectively enhanced by optimizing the learning objective.

5) Prediction Layer and Model Optimization:

a) Prediction layer: First, given the current session s , we use the corresponding local session representation $\mathbf{z}_s$ and augmented global session representation $\mathbf{z}_{g(s)}$ to compute the similarity between the session representations and the augmented global item embedding $\mathbf{h}_{g:i}^{b_1}$ of item v_i by taking their inner products. We then aggregate the two similarities to obtain the score ψ_i of each candidate item v_i . The method is as follows:

$$\psi_i = \mathbf{h}_{g:i}^{b_1} \mathbf{z}_{g(s)}^T + \text{LeakyRelu} \left(\mathbf{h}_{g:i}^{b_1} \mathbf{z}_s^T \right) \quad (19)$$

where the function $\text{LeakyRelu}(\cdot)$ with hyperparameter γ is used to introduce a nonlinear transformation and prevent the gradient from vanishing and $\mathbf{z}_{g(s)}^T$ and $\mathbf{z}_s^T$ are the corresponding transposes.

The *softmax* function is subsequently used to calculate the probability that each item will become the next target behavior item in the current session.

$$\hat{\mathbf{y}} = \text{softmax}(\boldsymbol{\psi}) \quad (20)$$

where the vectors $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_N)$ and $\boldsymbol{\psi} = (\psi_1, \psi_2, \dots, \psi_N)$.

b) Model optimization: The learning objective is modeled as a cross-entropy loss function, which is widely used in recommendation systems and is defined as

$$\mathcal{L}_r = - \sum_{i=1}^N y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \quad (21)$$

where $\mathbf{y} \in R^N$ denotes a one-hot encoding vector for the real target item in the current session and $y_i \in (0, 1)$ represents the value at the position of index i of the vector $\mathbf{y}$. If the real target item of the current session is item v_i , then $y_i = 1$ and the values at the remaining index positions of $\mathbf{y}$ are 0.

By combining the recommendation loss ($\mathcal{L}_r$) and the contrastive learning loss ($\mathcal{L}_{item}$, $\mathcal{L}_{lin}$) and minimizing $\mathcal{L}$ using *Adam*, high-quality session-based recommendations can be obtained.

$$\mathcal{L} = \mathcal{L}_r + \delta(\mathcal{L}_{item} + \mathcal{L}_{lin}) \quad (22)$$

where δ is a hyperparameter that controls the ratio of contrastive learning loss.

C. Time Complexity Analysis

The time cost of MHCL is mainly determined by the graph convolution and soft-attention mechanism. There are four graph convolutions in MHCL. The time cost of two global hypergraph convolutions is $4JNd$. The time cost of the local heterogeneous hypergraph convolution is mainly determined by the node-to-hyperedge and hyperedge-to-node aggregation overheads. The former directly uses weighted summation, and its time cost is $J_{loc}xd$. The latter uses a soft-attention mechanism to compute the aggregation weights and then weighted summation, and its time cost is $J_{loc}x^2d^2$. Thus, the time cost of heterogeneous hypergraph convolution is $J_{loc}xd + J_{loc}x^2d^2$. The time cost of the multi-behavioral line graph convolution is $2J^2d$. The time costs of the soft-attention mechanism used in local session representation and global session representation learning are $2(xd^2 + d^2 + 2xd)$ and $xd^2 + 2d^2 + 2xd$, respectively; hence, the time cost of the soft-attention mechanism is $(3x + 4)d^2 + 6xd$. Combining the above analyses, the total time cost of MHCL is $4JNd + J_{loc}xd + J_{loc}x^2d^2 + 2J^2d + (3x + 4)d^2 + 6xd$. Because $N, J \gg d$, $x^2d^2 \gg xd$ and $x^2d^2 \gg J_{loc}$, the time complexity of MHCL is $O(JN + x^2d^2 + J^2)$.

IV. EXPERIMENTS

A. Experimental Settings

1) *Dataset*: In this study, we evaluated the performance of MHCL on three public real datasets: JD-Computer (abbreviated as JD-Com)¹, JD-Application (abbreviated as JD-App)¹, and Taobao². JD-Com and JD-App, both from the shopping site JD.com, contain anonymous user interaction sessions with computers and home appliances, respectively, and include 10 different behavior types. The Taobao dataset contains user behavior data provided by the Taobao shopping website, with four types of behavior. The preprocessing method for the datasets is as follows: (1) In the three datasets, we treat purchase behavior as the target behavior. For JD-Com and JD-App, we retain add-to-cart behavior as auxiliary behavior and delete other behavior types. For Taobao, we retain click behavior as the auxiliary behavior and delete other behavior types. (2) We removed items with too few interactions in the target behavior and sessions with too few target behavior items. Additionally, we removed all auxiliary behavior items that occurred after the penultimate target behavior item in ascending order of timestamps for each session. (3) The dataset was expanded using a sequence splitting method. For example, a session $s = [v_1, v_2, \dots, v_m]$ is split into m subsessions (i.e., $[v_1, v_2, \dots, v_m]$, $[v_1, v_2, \dots, v_{m-1}]$, $\dots$, $[v_1, v_2]$, and $[v_1]$). The longest subsession of each session is included in the test set, i.e., $[v_1, v_2, \dots, v_m]$, and the last target behavior item v_m in the longest subsession is treated as the test set label. The remaining subsessions are included in the

TABLE I
STATISTICS ON DATASET INFORMATION

Dataset	JD-Com	JD-App	Taobao
Train	3813	44527	147455
Test	9152	63422	9516
Item	5244	12504	11960
Target	14494	118697	150200
Auxiliary	20183	147620	60979
Session avg. len.	3.7890	4.1991	22.1919
Target avg. len.	2.2053	2.3276	6.4080

training sets, with the label being the first target behavior item that follows the last auxiliary behavior item of the subsession in the original session. (4) We deleted sessions in the training set that had no target behavior items other than the labels.

The preprocessed dataset information is shown in Table I, which lists the number of sessions in the training and test sets, the number of items, interaction times of the target and auxiliary behaviors, the average length of all sessions, and the average length of the target behavior sequences.

2) *Evaluation Metrics*: To evaluate the performance of MHCL and all the baselines, we used three widely recognized evaluation metrics: Hit Rate (abbreviated as $H@k$), Mean Reciprocal Rank (abbreviated as $M@k$) and Normalized Discounted Cumulative Gain (abbreviated as $N@k$), where k (with values of $k = 10$ or 20) denotes the number of recommended candidates. A higher value of all three metrics indicates better performance.

3) *Baselines*: We compared various session recommendation models: the GNN-based model [14], [18], [25], [28], hypergraph-based model [29], [32], contrastive learning-based model [29], [47], and multi-behavior-based model [35], [36], [51].

SR-GNN [18] treats each session as a graph of one-to-one correlations between items, obtains item embeddings via GNNs, and uses the last item as a query vector in a soft-attention mechanism to obtain session embeddings.

S²-DHCN [29] employs hypergraph convolutional networks and constructs line graphs as contrastive signals to improve performance through contrastive learning.

COTREC [47] explores the internal and external item relevance of sessions through contrastive learning and cotraining.

MGIR [25] uses sequential, co-occurrence, and incompatibility relations to model session graphs and extract finer granularity item correlations.

CLHHN [32] designs a category-aware lossless heterogeneous hypergraph neural network that uses the categories of items to recommend possible items.

MSGAT [28] is a multisparse graph attention network that uses two parallel channels to model intra- and inter-session item correlations.

GIUA-GNN [14] uses a global session graph and bidirectional attention graph convolutional network to learn user and item embeddings.

GPG4HSR [36] introduces auxiliary behavior sequences and uses global and personalized graphs to learn behavioral transitions and user intentions.

¹<https://drive.google.com/drive/folders/1kz7Dkq5jQ8WR82xBITyQ7S4vUgBFNsDn>

²<https://tianchi.aliyun.com/dataset/649>

TABLE II
PARAMETER SETTINGS OF MHCL

Dataset	d	$batch$	$length$	lr	L_g	L_s	μ	δ
JD-Com	100	128	50	$1e^{-2}$	3	2	0.5	$5e^{-3}$
JD-App	100	128	50	$5e^{-5}$	3	2	0.5	$1e^{-3}$
Taobao	100	128	50	$5e^{-5}$	3	2	0.3	$1e^{-3}$

MB-STR [51] captures item heterogeneous dependency via multihead self-attention and designs a multi-behavior sequential pattern generator to capture item sequential dependency.

BA-GNN [35] considers the effects of different behaviors and constructs directed graphs for the sequences of two interaction behaviors. It aggregates representations of the sequences of different behaviors using a gating mechanism.

4) *Parameter Settings*: The parameter settings for MHCL are listed in Table II. The optimal values for the learning rate (lr), the number of convolution layers (L_g and L_s), μ and δ were determined by grid search. For the baselines, the session sequence length was set to 50 for a fair comparison. However, because the BA-GNN performs poorly with a session length of 50, we adjusted its length using a grid search to select the optimal length. Specifically, we set the length to 3 for JD-Com and JD-App and 5 for Taobao. For the other hyperparameters of the baselines, we referred to the best parameter settings reported in the original baseline papers and then fine-tuned the hyperparameters to ensure that the baselines achieved optimal performance.

B. Analysis of Experimental Results

1) *Overall Performance*: The overall performance of all the models on the three datasets is listed in Table III, where the best results are shown in bold and the second-best results are underlined. Compared with the other baselines, both BA-GNN, MB-STR and GPG4HSR introduced auxiliary behaviors and achieved better results. Therefore, it can be concluded that a reasonable introduction of auxiliary behaviors can improve recommendation performance. The results of GPG4HSR and MB-STR are generally better than those of BA-GNN because they model the item correlations between different behaviors, preserve the positional information between items with different behaviors, and align more closely with realistic perspective modeling, thereby extracting user preferences more accurately.

S²-DHCN achieved better results than the baselines on the Taobao dataset and yielded good results on the JD-Com and JD-App datasets. This is due to the advantages of hypergraph modeling; the higher-order correlations of items extracted by the hypergraph can better represent item relationships and are more realistic. Additionally, the performance of BA-GNN and GPG4HSR on Taobao is worse than that on the other two datasets compared with the baseline method of single behavior because the auxiliary behaviors in the Taobao dataset contain more noisy information, leading to a certain bias in the extracted user preferences. The larger the noise ratio is, the more noise information there is in the auxiliary behavior. The calculated noise ratios for auxiliary behaviors on the JD-Com, JD-App, and Taobao datasets are 0.4826, 0.4016, and 0.9056, respectively, indicating

that the Taobao dataset contains more noise information in its auxiliary behaviors. CLHHN also achieved good results among the baselines because of the introduction of item attribute information and more granular modeling of user preferences. The introduction of additional auxiliary information is beneficial for alleviating sparsity and effectively improving recommendation performance. Furthermore, the performance advantages exhibited by S²-DHCN and COTREC demonstrate that contrastive learning is effective in enhancing user preference representation.

MHCL achieves the best performance on all datasets for the following reasons: (1) Using the local session heterogeneous hypergraph constructed, the relative positions of items between different behaviors are preserved. Thus higher-order correlations of items between different behaviors are effectively captured in the current session. (2) Higher-order correlations of items are modeled using hypergraphs, which provide a more realistic representation. (3) The introduction of contrastive learning helps to effectively use the interactive information under the auxiliary behavior, thus effectively enhancing the global item embedding. (4) An efficient contrastive signal is designed for global session representation to learn user preferences more accurately. (5) Personalized weights are set for items with different behaviors during local heterogeneous hypergraph convolution. Additionally, a soft-attention mechanism with two different query functions is adopted when local item embeddings are aggregated, helping to filter out noise information in auxiliary behaviors to a certain extent. Consequently, MHCL can provide more accurate personalized recommendations.

2) *Ablation Experiment*: To explore the necessity of key components in MHCL and their impact on the overall performance, four variants of MHCL were designed, and ablation experiments were conducted on three datasets. The four variants are as follows: MHCL_{nm1} (i.e., no multi-behavior line graphs or their corresponding contrastive learning modules were used.), MHCL_{ncl} (i.e., the contrastive learning module is not used in global item-embedding learning), MHCL_{n1h} (i.e., the local session heterogeneous hypergraph is replaced by modeling each behavior sequence separately; the target and auxiliary behavior sequences are mean pooled separately, and a gating mechanism is then used to generate the local session representation), and MHCL_{ng} (i.e., only local session representations are used; the global session representation is removed when calculating candidate item scores).

The results of the ablation experiments are presented in Table IV. The performance of the model decreased when any key component was removed, with the exception that N@10 and N@20 for MHCL_{ncl} on JD-App slightly improved. We hypothesize that the global item embedding based on auxiliary behavior may learn noise, which slightly negatively impacts the final results through contrastive learning. Compared with MHCL_{nm1}, MHCL uses a novel multi-behavior line graph as a contrastive signal for contrastive learning, which can extract higher-order correlations of items between different behaviors in the global view and effectively enhance the global session representation. Compared with MHCL_{ncl}, contrastive learning can enhance the rich item information in the auxiliary behavior into the global item embedding, which generates a more reliable

TABLE III
COMPARISON OF OVERALL PERFORMANCE OF MHCL AND ALL BASELINE MODELS

Dataset	Models	H@10	M@10	N@10	H@20	M@20	N@20
JD-Com	SR-GNN	2.79	1.05	-	4.98	1.20	-
	S ² -DHCN	4.29	1.26	1.97	6.48	1.42	2.52
	COTREC	4.10	1.11	1.81	5.91	1.24	2.27
	MGIR	1.86	0.77	1.01	2.93	0.84	1.27
	CLHHN	4.37	1.01	1.79	8.02	1.25	2.69
	MSGAT	0.88	0.45	0.55	1.22	0.47	0.63
	GIUA-GNN	3.45	1.31	1.80	5.43	1.44	2.30
	GPG4HSR	5.33	2.05	<u>2.79</u>	7.63	2.20	3.36
	MB-STR	<u>5.94</u>	1.71	2.63	<u>9.47</u>	1.82	3.33
	BA-GNN	5.11	<u>2.11</u>	2.78	7.53	<u>2.28</u>	<u>3.41</u>
	MHCL	7.83	2.71	3.76	11.07	2.97	4.52
	Imp. (%)	31.82	28.44	34.77	16.90	30.26	32.55
JD-App	SR-GNN	3.80	1.31	-	6.31	1.48	-
	S ² -DHCN	11.22	3.34	5.19	15.36	3.63	6.24
	COTREC	10.79	2.92	4.76	15.02	3.20	5.82
	MGIR	2.61	0.76	1.18	4.16	0.56	1.58
	CLHHN	<u>13.02</u>	3.55	<u>5.77</u>	<u>17.57</u>	3.93	<u>7.17</u>
	MSGAT	0.98	0.83	0.87	1.12	0.84	0.90
	GIUA-GNN	6.36	2.97	3.76	8.65	3.13	4.34
	GPG4HSR	11.12	4.09	5.71	15.04	4.35	6.70
	MB-STR	10.36	<u>4.19</u>	5.64	14.17	<u>4.45</u>	6.59
	BA-GNN	10.05	3.87	5.31	14.49	4.18	6.43
	MHCL	13.05	4.91	6.18	17.70	4.99	7.27
	Imp. (%)	0.23	17.18	7.11	0.74	12.13	1.39
Taobao	SR-GNN	3.18	1.19	-	5.03	1.31	-
	S ² -DHCN	<u>5.57</u>	1.33	<u>2.30</u>	8.59	1.53	<u>3.05</u>
	COTREC	5.11	1.29	2.18	7.77	1.47	2.84
	MGIR	1.23	0.44	0.62	2.13	0.51	0.85
	CLHHN	2.87	1.11	1.53	3.96	1.19	1.80
	MSGAT	0.09	0.03	0.04	0.20	0.03	0.06
	GIUA-GNN	4.60	<u>1.60</u>	2.29	7.40	<u>1.79</u>	3.00
	GPG4HSR	4.93	1.27	2.07	7.98	1.27	2.84
	MB-STR	5.42	1.30	2.19	<u>8.94</u>	1.41	2.89
	BA-GNN	4.07	1.35	1.97	6.58	1.51	2.60
	MHCL	6.20	1.88	2.77	9.32	2.06	3.47
	Imp. (%)	11.31	17.5	20.43	4.25	15.08	13.77

global item embedding. Compared with MHCL_{nlh}, the effectiveness of modeling the current session using the heterogeneous hypergraphs proposed in this study was verified. This approach preserves the relative positions between items with different behaviors in the original session and models the higher-order correlations between items with different behaviors, which is better than modeling the target behavior sequence and the auxiliary behavior sequence separately. Compared with MHCL_{ng}, modeling with all sessions can obtain richer information from other sessions.

3) *Hyperparameters*: In this section, we investigate the effects of the following hyperparameters on the performance of MHCL: the number of layers L_g of the global hypergraph, μ , and δ , which controls the percentage of contrastive learning loss.

a) *Impact of L_g* : The number of layers in the global hypergraph convolution indicates the maximum number of orders of neighbors from which information can be obtained. We adjusted

L_g in the range $\{1, 2, 3, 4, 5, 6\}$. As shown in Fig. 4, the optimal results are mostly achieved with three layers. Although the performance varied across datasets, a common feature was that they all yielded poorer results when the number of layers was one. This finding indicates that using only one convolution layer is not sufficient to obtain more comprehensive higher-order correlations of items. An appropriate number of layers allows less redundant information to propagate during convolution while capturing more comprehensive higher-order correlations of items.

b) *Impact of μ* : The hyperparameter μ controls how much information is propagated to the nodes by their neighbors from the auxiliary behavior edges during the convolution of the multi-behavior line graph. μ was adjusted within the range of $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$. The larger the value is, the more information is propagated. Fig. 5 shows that most of the optimal results for the JD-Com and JD-App datasets are obtained at

TABLE IV
THE RESULTS OF THE MHCL ABLATION EXPERIMENTS (THE BEST RESULTS ARE SHOWN IN BOLD)

Dataset	Metric	MHCL _{lml}	MHCL _{lcl}	MHCL _{lnh}	MHCL _{lng}	MHCL
JD-Com	H@10	6.61	6.88	7.41	5.89	7.83
	M@10	2.04	2.51	2.70	2.16	2.71
	N@10	3.08	3.53	3.66	3.03	3.76
	H@20	9.58	9.46	10.55	8.45	11.07
	M@20	2.21	2.69	2.86	2.34	2.97
	N@20	3.82	4.18	4.36	3.67	4.52
JD-App	H@10	12.05	12.59	12.73	11.12	13.05
	M@10	3.45	4.11	3.95	4.36	4.91
	N@10	5.47	6.25	5.97	5.94	6.18
	H@20	17.05	16.87	17.26	15.29	17.70
	M@20	3.80	4.38	4.25	4.64	4.99
	N@20	6.73	7.30	7.11	6.99	7.27
Taobao	H@10	5.78	5.02	6.07	4.47	6.20
	M@10	1.70	1.06	1.87	1.40	1.88
	N@10	2.58	1.96	2.76	2.10	2.77
	H@20	8.59	8.52	9.11	7.32	9.32
	M@20	1.91	1.30	1.97	1.60	2.06
	N@20	3.33	2.84	3.40	2.82	3.47

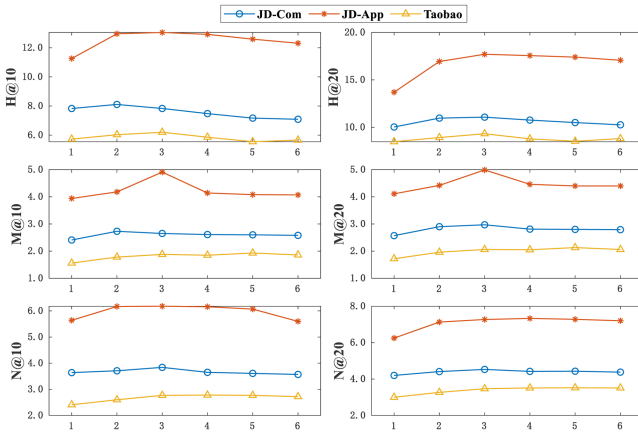


Fig. 4. Impact of L_g .

$\mu = 0.5$, whereas most of the optimal results for the Taobao dataset are obtained at $\mu = 0.3$.

c) *Impact of δ* : The hyperparameter δ controls the percentage of contrastive loss and is adjusted in the range of $\{0.01, 0.005, 0.001, 0.0005, 0.0001\}$, with larger δ values resulting in a larger percentage of contrastive loss. As shown in Fig. 6, for JD-Com, the best results are obtained at $\delta = 0.005$, with the optimal results for H@10 and N@10 obtained at $\delta = 0.001$. For JD-App, the optimal results for H@10, H@20, and N@20 are achieved at $\delta = 0.001$, with the overall trend increasing to the optimum and then decreasing as δ decreases. The optimal results for M@10, M@20, and N@20 are achieved at $\delta = 0.0001$. For Taobao, the optimal results are obtained at $\delta = 0.001$, with the overall trend showing an increase to the optimum and then a decrease as δ decreases.

4) *Comparison of Runtimes*: The experiment was run on RTX3060, and the runtime of an epoch and the total runtime required to converge to the optimum for all the models are shown in Table V. CLHHN, GPG4HSR, MB-STR, BA-GNN,

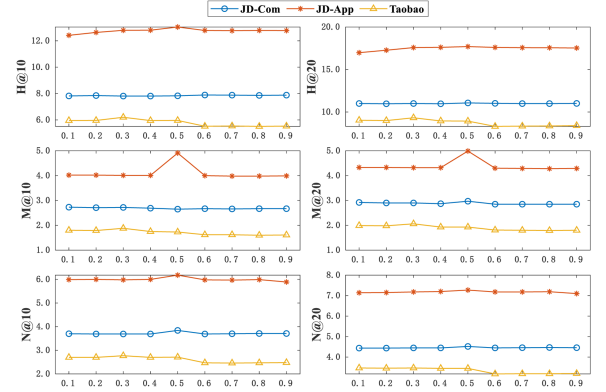


Fig. 5. Impact of μ .

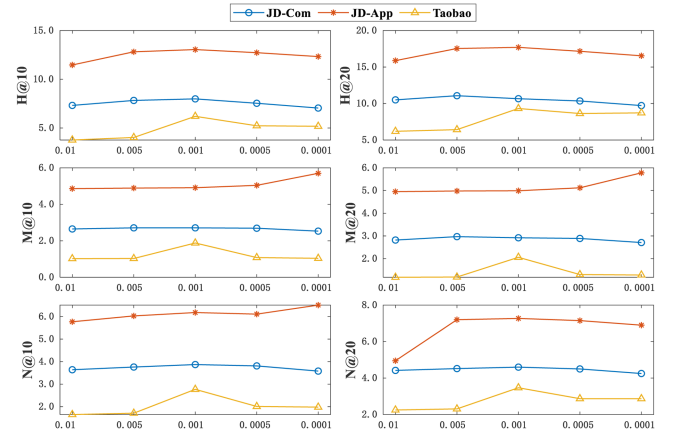


Fig. 6. Impact of δ .

TABLE V
COMPARISON OF RUNTIMES

Models	JD-Com (s)		JD-App (s)		Taobao (s)	
	one epoch	total	one epoch	total	one epoch	total
SR-GNN	3	36	37	407	41	328
S ² -DHCN	11	77	81	810	49	343
COTREC	3	21	42	378	30	270
MGIR	1	18	12	252	15	225
CLHHN	13	143	110	1320	62	558
MSGAT	5	70	57	912	55	1045
GIUA-GNN	4	64	42	462	59	472
GPG4HSR	12	168	138	1656	227	2497
MB-STR	8	208	98	2254	134	3752
BA-GNN	36	1512	39	1209	235	12455
MHCL	33	759	448	8512	202	4242

and MHCL with auxiliary behavior are more time consuming than the other baselines.

In addition, we further analyze the time complexity of several multi-behavior baselines. The time cost of GPG4HSR is determined mainly by the global graph convolution, personalized graph convolution, and self-attention layers. The time cost of the global graph convolution is $6N^2d + 28Nd$, that of the personalized graph convolution is $2N^2d$, and that of the self-attention layer is $x^2d + xd^2$. Therefore, total time cost of GPG4HSR is $8N^2d + 28Nd + x^2d + xd^2$, and its time complexity is

$O(N^2)$. The time cost of MB-STR is mainly determined by the multi-behavior transformer layer and is $O(x^2d + xd^2)$. The time cost of BA-GNN is determined mainly by the two gated graph convolutions and the self-attention layer. The time cost of two gated graph convolutions is $(N^2 + 7)xd + 2xd^2$, and that of the self-attention layer is $x^2d + 3xd^2$. Therefore, its total time cost is $2N^2xd + 14xd + x^2d + 7xd^2$, and its time complexity is $O(N^2)$. Their time costs are mainly caused by graph convolution and attention mechanism.

The relatively high time cost of MHCL is due to the multiple hypergraphs as well as the attention mechanism that adds additional parameters, thus increasing the time cost. However, MHCL provides a significant improvement in recommendation accuracy over other baselines, even if the time cost is worth it.

V. CONCLUSION

This paper proposed a novel multi-behavior hypergraph contrastive learning model for session recommendation. Auxiliary behaviors are introduced and effective item embeddings are learned to extract user preferences more comprehensively from both local and global perspectives. A local session heterogeneous hypergraph is designed in the local session representation module to adequately model higher-order correlations of items while preserving the relative positions between items with different behaviors. In the global session representation module, a multi-behavior line graph is constructed as an effective contrastive signal to further capture the higher-order correlations of items between behaviors from the global perspective. Extensive experiments proved that MHCL can effectively improve the recommendation accuracy.

The MHCL model also has several limitations: (1) it uses only one auxiliary behavior type and does not model all auxiliary behavior types; (2) the computational complexity of the model is relatively high. Therefore, in future work, we will consider modeling more types of auxiliary behaviors and attempt to design lighter session-based recommendation models. Additionally, owing to the sparse nature of session data, we will attempt to introduce more auxiliary information, such as item attributes and time interval information between items, to improve session recommendation performance.

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